MIL-A-45062E (MU)
16 July 1973
SUPERSEDING
MIL-A-45062D (MU)
14 May 1965

MILITARY SPECIFICATION

ACTIVATOR, M2
LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of high explosive activator designated as Activator, M2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-45486 - Pellets RDX.
MIL-D-46653 - Detonators, Loading, Assembling and Packing
MIL-A-2550 - Ammunition and Special Weapons, General Specification for.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).
MIL-STD-109 - Quality Assurance Terms and Definitions.
MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

FSC: 1345

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DRAWINGS

MUNITIONS COMMAND

7548048 - Activator, AT, Mine, M2.
7548390 - Box, Packing, Ammunition for Activator M1; Activator Prac., M1 and M2 HE Activator Assembly Details, and Packing.
7548391 - Ammunition Container, Metal for Activator M1; Activator, Prac., M1 and Activator M2, Assembly Details, Packing and Marking.
8796522 - Marking Diagram and Sealing for Wood Packing Boxes.
8873797 - Test Fixture.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other Publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Code of Federal Regulations, Title 49 - CFR 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders shall specify 49 CFR 100-199 [latest revision]).

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Activators.-The activator shall comply with all requirements specified on drawing (dwg.) 7548048 and with all requirements specified in applicable specifications. RDX pellets shall be Type II as specified in MIL-P-45486 (Ord).

3.3 Jolt and Jumble.-The loaded activators shall withstand the jolt test specified in Standard MIL-STD-331, Test No. 101.1. After completion of the jolt test, the same activators shall with-

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stand the jumble test specified in Standard MIL-STD-331, Test No. 102.1. In addition, there shall be no evidence of cracks, broken edges, separation of parts, loosening of seals, or exposed explosive.

3.4 Functioning.-The activator shall function satisfactorily as evidenced by destruction of one half, minimum (min.) of an aluminum cylinder, 11/16 inch diameter by 1 5/8 inch long.

3.5 Marking.-Marking shall be restored by the loading and assembly facility on any components furnished to it marked when such markings have been marred or obliterated.

3.6 Workmanship.-All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, rust, chips, grease, corrosion, and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents or the explosives involved. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material.

3.7 First article inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of Product.-At the time each completed lot of items deliverable under the contract is submitted to the

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Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted.

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specification referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agents authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission.-Prior to the start of regular production the contractor shall submit a first article sample (see 6.1c) to a Government approved facility as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2 and 4.2.3. The first article sample shall consist of 70 loaded activator assemblies which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials including packaging and packing shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including a certificate of conformance for materials. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90

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days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 First article inspection provisions for loaded activator assemblies.-Seventy (70) loaded activator assemblies completely packaged and sealed as required by the applicable drawings shall be submitted and subjected to the following tests:

a. Jolt and jumble (see 3.3) - 35 assemblies, in accordance with the method specified in 4.3.3.1.
b. Functioning (see 3.4) - 35 assemblies, in accordance with the method specified in 4.3.3.2.

4.2.3. Rejection.-The first article sample shall be rejected if:

a. A critical defect occurs (see 3.3 and 4.3.3.1).
b. Two or more assemblies fail to meet the functioning requirement (see 3.4 and 4.3.3.2).

The Government reserves the right to terminate its inspection upon any failure of an item in the sample to comply with any of the stated requirements. Until that phase of first article samples consisting of loaded items is accepted, the contractor is in no way authorized by the Government to initiate regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process using the same materials and methods in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specifications, or drawings not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not

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be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of activator assemblies shall contain:

a. Parts from one manufacturer.
b. Detonators from not more than one lot from one manufacturer.
c. Pellets, RDX from not more than one lot from one manufacturer.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraphs of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Pellet (see dwg. 7548053 covering a detail of dwg. 7548048).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical: None defined.			
Major: None defined.			
Minor:	AQL 1.5 percent		
201.	Length, minimum (min.).....	Gage	01001
202.	Diameter.....	Gage	01002
203.	Pellent cracked, chipped or broken.....	Visual	01003

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204.	Evidence of poor workmanship (see 3.6).....	Visual	01004

4.3.2.2 Assembly (prior to cementing the cup and assembling the plug) (see dwg. 7548048).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Detonator assembly inverted or missing.....	Visual	02001
102.	Pellet missing.....	Visual	02002
103.	Flash hole obstructed.....	Visual	02003
104.	Evidence of poor workmanship on detonator, in flash hole, or plug threads, (see 3.6).....	Visual	02004
Minor:	AQL 1.5 percent		
201.	Detonator assembly insecure (cement inadequate.....	Visual-Manual	02005
202.	Cushion not completely secured to body (adhesive inadequate).....	Visual-Manual-	02006
203.	Sensitive end (red end) of detonator coated with cement.....	Visual	02007
204.	Onion skin disc missing or punctured.....	Visual	02008

4.3.2.3 Assembly (see dwg. 7548048).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 percent		
101.	True positioning of body thread with diameter of cup.....	Gage	03001
102.	Cup loose or not fully seated.....	Visual-Manual	03002
103.	Plug missing or damaged.....	Visual	03003
104.	Gasket missing.....	Visual	03004
105.	Assembly damaged (crack, split, or deep scratches).....	Visual	03005
106.	Body thread damaged.....	Visual	03006

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Minor:	AQL 2.5 percent		
201.	Plug turns when min torque is applied in loosening direction.....	Test	03007
202.	Total length, maximum (max.).....	Gage	03008
203.	Length from bottom of cup to gasket, max.....	Gage	03009
204.	Marking misleading or unidentifiable.....	Visual	03010
205.	Evidence of poor workmanship (see 3.6).....	Visual	03011

4.3.2.4 Sealed metal container (see dwg. 7548391).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Contents move when shaken.....	Manual	04001
102.	Container dented, punctured, split, or with seam separating.....	Visual	04002
Minor:	AQL 1.0 percent		
201.	Bare spot on exterior other than slight scratches.....	Visual	04003
202.	Marking misleading or unidentifiable.....	Visual	04004

4.3.2.5 Box, packing, wooden, sealed (see dwg. 7548390).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Hardware missing, broken or loose.....	Visual-Manual	05001
102.	Strapping missing, broken or loose.....	Visual-Manual	05002
103.	Contents exposed.....	Visual	05003
104.	Board broken or split.....	Visual	05004

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Minor:	AQL 2.5 percent		
201.	Hardware improperly engaged.....	Visual-Manual	05005
202.	Handle missing or insecure.....	Visual-Manual	05006
203.	Contents loose when box is tilted..	Manual	05007
204.	Strapping improperly assembled.....	Visual	05008
205.	Car seal missing unsealed or improperly positioned.....	Visual	05009
206.	Marking misleading or unidentifiable.....	Visual	05010

4.3.3 Testing

4.3.3.1 Jolt and Jumble (see 3.3) Code No. 03012 - Critical defect if any detonator functions; Major defect for all others.- Twelve (12) loaded activators shall be selected from each lot for these tests. The tests shall be performed as specified in 4.4.1 using equipment in accordance with 4.3.4. If one or more activators fail to comply with the specified requirements, the lot shall be rejected.

4.3.3.2 Functioning (see 3.4) Code No. 03013 - Major defect.- Fifty (50) loaded activators shall be selected from each lot for this test. If 2 or more activators fail to comply with the requirement specified, the lot shall be rejected. If only one defective is found, a second sample of 100 activators shall be tested. If the combined number of defectives in both the first and second samples is two or more, the lot shall be rejected. The test shall be performed as specified in 4.4.2 using equipment in accordance with 4.3.4.

4.3.3.3 Check test for deterioration of detonators (see 06001 specification MIL-D-46653) Major defect.-If the total time between original acceptance of any detonator lot and the assembly of that lot into the activators exceed 2 years, or if the detonators have been subjected to adverse conditions, however brief, at any time since previous tests, the detonator lot shall be subjected to, and must satisfactorily pass, the check test specified in the applicable detonator loading specification immediately before the detonator lot is assembled into the activators. This test shall be performed by the contractor on detonators selected by the Government inspector at the facility assembling the detonators into the activators. The test shall be conducted at Government expense without cost to the contractor who loaded the detonator or the contractor assembling the detonators into the activators (see 6.4) and shall not constitute

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a basis of rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc. permitted while the detonator lot was under the jurisdiction of either contractor.

4.3.3.4 RDX Pellets (see dwg. 7548053) Major defect.- Sampling, acceptance criteria and test procedures for this test shall be as specified in MIL-P-45486 (MU).

4.3.4 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" headings in 4.3.2 and in the procedures in 4.4 (see 6.3).

4.4 Test Methods and Procedures

4.4.1 Jolt and Jumble.-The loaded activators shall be assembled to a specified adapter and subjected to the jolt test in accordance with MIL-STD-331, Test 101.1. After completion of the Jolt test, the same activators shall be removed from the adapter and subjected to the Jumble test in accordance with MIL-STD-331, Test 102.1. Dis-assembly of the activators for examination shall be accomplished after both jolt and jumble tests have been completed. Examination shall be made to determine compliance with 3.3.

4.4.2 Functioning.-The activators shall be immersed in water to a depth of six inches, minimum, at ambient temperature for not less than 48 hours. Within one half hour after removal from the water, the activators shall be assembled into inspection equipment as specified on dwg. 8873797, and shall contact the top of a preweighed aluminum cylinder, 11/16 inch diameter by 1 5/8 inch long, and fired. Observation shall be made to determine compliance with 3.4 as evidenced by destruction of at least one half of the aluminum cylinder as determined by weighing the remaining portion.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A.-The loaded Activators, M2 shall be preserved and packaged in accordance with dwg. 7548391.

5.1.2 Level C.-The loaded activators, M2 shall be packaged

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and packed in accordance with requirements of the Code of Federal Regulations, 49 CFR 100-199.

5.2 Packing

5.2.1 Level A.-The loaded Activators, M2 packaged as specified in 5.1.1 shall be packed in accordance with dwg. 7548390.

5.2.2 Level C.-Packing shall be as specified in Paragraph 5.1.2.

5.3 Marking, Level A.-Marking shall be in accordance with dwgs. 7548390 and 7548391.

5.3.1 Marking, Level C.-Marking shall be in accordance with dwg. 8796522.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. They should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.3).

6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

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6.3.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public.) Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of Designs for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Picatinny Arsenal, ATTN: SARPA-QA-A-M, Dover, New Jersey 07801 in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion

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date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Cost of check test.-The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test for deterioration of the detonators. The test shall be conducted at Government expense without cost to the contractor assembling the detonators into the activators and shall not constitute a basis for rejection against the contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the detonator lot was under the jurisdiction of the contractor.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project No. 1345-A-026

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